

No-Free-Lunch Theorems: Three Variants

Natural-Language Proofs

Proof Agent

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1 Overview

We present natural-language proofs for three variants of the No-Free-Lunch (NFL) theorem. All three capture the same meta-theorem in different settings:

1. **Online adversarial NFL (Theorem 2.1):** no deterministic online learner can avoid mistakes against a worst-case Boolean target on any pre-fixed query sequence.
2. **Wolpert–Macready histogram NFL (Theorem 3.1):** averaged over all cost functions, any two non-revisiting stochastic search algorithms see identical observation-sequence distributions.
3. **Batch stochastic NFL (Theorem 4.1):** for any learning algorithm on a finite domain, a realizable distribution exists under which the algorithm fails with constant probability.

Throughout, X is the *domain*, Y is the *codomain* (the range of target / cost functions), and $\{0, 1\}$ denotes the set of Boolean values $\{0, 1\}$ (written as `Bool` in Lean 4).

2 Theorem 1: Online Adversarial NFL

2.1 Setting

A *deterministic online learner* A is a function

$$A : \text{List}(X \times \{0, 1\}) \rightarrow X \rightarrow \{0, 1\}$$

that receives the history of previously seen (query, label) pairs and then predicts the label of a new query point.

Given a target $f : X \rightarrow \{0, 1\}$ and a query sequence $xs = [x_0, x_1, \dots, x_{n-1}]$, define the *history at round t* :

$$\text{hist}(f, xs, t) = [(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_{t-1}, f(x_{t-1}))]$$

The learner *makes a mistake at round t* (written $\text{MistakeAt}(A, f, xs, t)$) if

$$A(\text{hist}(f, xs, t))(x_t) \neq f(x_t) \quad \text{whenever } t < |xs|.$$

2.2 Statement

Theorem 2.1 (Adversarial Online NFL). *For any deterministic online learner A and any duplicate-free (nodup) query sequence xs , there exists a target function $f : X \rightarrow \{0, 1\}$ such that A makes a mistake at every round t (i.e., $\text{MistakeAt}(A, f, xs, t)$ holds for all t).*

2.3 Proof

By induction on the list xs .

Base case ($xs = []$). The empty sequence has no rounds; any target works vacuously.

Inductive step ($xs = x :: xs'$). Assume the theorem holds for every algorithm on the tail xs' . Since xs is duplicate-free, $x \notin xs'$ and xs' is itself duplicate-free.

Construction.

1. Let $b = \neg A([], x)$, the opposite of A 's prediction at x when it has seen no history.
2. Define the *shifted learner* $A'(h, y) = A((x, b) :: h, y)$. Intuitively, A' is A with (x, b) prepended to every history.
3. Apply the induction hypothesis to A' and xs' : there exists $g : X \rightarrow \{0, 1\}$ such that A' makes a mistake at every round of xs' against g .
4. Define the target:

$$f(y) = \begin{cases} b & \text{if } y = x \\ g(y) & \text{otherwise.} \end{cases}$$

Verification that A mistakes at every round of $x :: xs'$ against f .

Round $t = 0$ (query x). The history is empty. A 's prediction is $A([])(x)$. By definition $f(x) = b = \neg A([])(x)$, so the prediction differs from the true label.

Round $t = s + 1$ (query $xs'[s]$). Since $xs'[s] \in xs'$ and $x \notin xs'$, we have $xs'[s] \neq x$, so $f(xs'[s]) = g(xs'[s])$.

The history that A sees at round $s + 1$ when run on $x :: xs'$ against f is:

$$[(x, f(x)), (xs'[0], f(xs'[0])), \dots, (xs'[s-1], f(xs'[s-1]))].$$

Because $xs'[i] \neq x$ for all i , we have $f(xs'[i]) = g(xs'[i])$ for each i . Thus the history equals

$$(x, b) :: [(xs'[0], g(xs'[0])), \dots, (xs'[s-1], g(xs'[s-1]))] = (x, b) :: \text{hist}(g, xs', s).$$

Therefore:

$$A(\text{hist}(f, x :: xs', s+1))(xs'[s]) = A((x, b) :: \text{hist}(g, xs', s))(xs'[s]) = A'(\text{hist}(g, xs', s))(xs'[s]).$$

By the induction hypothesis, A' mistakes at round s of xs' against g , meaning $A'(\text{hist}(g, xs', s))(xs'[s]) \neq g(xs'[s]) = f(xs'[s])$. $\square$

Remark. The constructed target f is built *adaptively*: it first sets the label at x to fool A at round 0, then delegates to g (which was designed to fool the shifted algorithm A' on the remaining queries). The non-repetition condition (xs is nodup) is essential: if x were queried again, the consistency of f could not be maintained.

3 Theorem 2: Wolpert–Macready (Histogram Form)

3.1 Setting

Let X and Y be finite types. A *search algorithm* A maps a history $h \in \text{List}(X \times Y)$ to a probability mass function (PMF) over X . The algorithm is *non-revisiting* if it assigns zero probability to any point already appearing in the history:

$$x \in h.\text{keys} \Rightarrow A.\text{nextQuery}(h)(x) = 0.$$

Given a target cost function $f : X \rightarrow Y$, the *m-step observation PMF* of A on f is the distribution over length- m sequences of observed costs $c \in Y^m$ produced when A queries f for m steps (with labels revealed after each query).

3.2 Statement

Theorem 3.1 (Wolpert–Macready, histogram form). *For any two non-revisiting search algorithms A_1, A_2 , finite domain X , finite codomain Y , budget $m \leq |X|$, and any observation sequence $c \in Y^m$:*

$$\sum_{f: X \rightarrow Y} \Pr[A_1 \text{ produces } c \text{ on } f] = \sum_{f: X \rightarrow Y} \Pr[A_2 \text{ produces } c \text{ on } f].$$

3.3 Proof

Key object. For a non-revisiting algorithm A , a history h whose query keys are distinct, and a future observation sequence c , define:

$$\text{avgFuture}(A, h, c) = \sum_{\substack{f: X \rightarrow Y \\ f \text{ consistent with } h}} \Pr[A \text{ produces } c \text{ from history } h \text{ on } f],$$

where f is *consistent with* h if $f(x) = y$ for every $(x, y) \in h$.

Lemma 3.2 (Algorithm-independent value). *If A is non-revisiting, h has distinct query keys, and $|h| + |c| \leq |X|$, then:*

$$\text{avgFuture}(A, h, c) = |Y|^{|X| - |h| - |c|}.$$

Proof. By induction on c .

Base case ($c = []$). The empty observation sequence is produced with probability 1 by every consistent target (there are no steps to take). Hence:

$$\text{avgFuture}(A, h, []) = |\{f : X \rightarrow Y \mid f \text{ consistent with } h\}|.$$

A consistent function must agree with h on the $|h|$ queried points and is free on the remaining $|X| - |h|$ points. Therefore the count is $|Y|^{|X| - |h|}$, which equals $|Y|^{|X| - |h| - 0}$. ✓

Inductive step ($c = y :: ys$). At each step, the algorithm draws the next query x from the PMF $A.\text{nextQuery}(h)$. Only if $f(x) = y$ does x contribute to the observation sequence. Expanding:

$$\begin{aligned} \text{avgFuture}(A, h, y :: ys) &= \sum_{\substack{f \text{ consistent with } h \\ x \in X}} \sum_{x \in X} A.\text{nextQuery}(h)(x) \cdot \mathbf{Pr}[A \text{ first observes } y \text{ then } ys \mid \text{next query is } x, f] \\ &= \sum_{x \in X} A.\text{nextQuery}(h)(x) \cdot \sum_{\substack{f \text{ consistent with } h \\ f(x)=y}} \mathbf{Pr}[A \text{ produces } ys \text{ from } h \cup \{(x, y)\} \text{ on } f]. \end{aligned}$$

Two observations justify this rearrangement:

- The non-revisiting condition ensures $A.\text{nextQuery}(h)(x) = 0$ whenever $x \in h.\text{keys}$. Thus only fresh x contribute.
- When $x \notin h.\text{keys}$, the set of functions consistent with h and satisfying $f(x) = y$ is exactly the set of functions consistent with $h \cup \{(x, y)\}$.

Recognizing the inner double sum as $\text{avgFuture}(A, h \cup \{(x, y)\}, ys)$:

$$\text{avgFuture}(A, h, y :: ys) = \sum_{x \in X} A.\text{nextQuery}(h)(x) \cdot \text{avgFuture}(A, h \cup \{(x, y)\}, ys).$$

By the induction hypothesis, $\text{avgFuture}(A, h \cup \{(x, y)\}, ys) = |Y|^{|X| - (|h|+1) - |ys|} = |Y|^{|X| - |h| - |c|}$ (a constant independent of x). Since $A.\text{nextQuery}(h)$ is a probability distribution summing to 1:

$$\text{avgFuture}(A, h, y :: ys) = |Y|^{|X| - |h| - |c|} \cdot \underbrace{\sum_{x \in X} A.\text{nextQuery}(h)(x)}_{=1} = |Y|^{|X| - |h| - |c|}. \quad \checkmark$$

□

Conclusion of Theorem 3.1. Apply Lemma 3.2 with $h = []$ (empty history, trivially consistent with every target) and the given c of length $m \leq |X|$. Every target is consistent with the empty history, so:

$$\sum_{f: X \rightarrow Y} \mathbf{Pr}[A_i \text{ produces } c \text{ on } f] = \text{avgFuture}(A_i, [], c) = |Y|^{|X| - m}.$$

This value is the same for both A_1 and A_2 . □

Corollary 3.3 (Averaged performance). *Under the same hypotheses, for any score function $\varphi : Y^* \rightarrow \mathbb{R}_{\geq 0}$:*

$$\sum_{f: X \rightarrow Y} \mathbf{E}[\varphi(A_1 \text{ on } f)] = \sum_{f: X \rightarrow Y} \mathbf{E}[\varphi(A_2 \text{ on } f)].$$

Proof. Expand each expectation as a sum over observation sequences c and swap the two sums:

$$\begin{aligned}\sum_f \mathbf{E}[\varphi(A_i \text{ on } f)] &= \sum_f \sum_{c \in Y^*} \mathbf{Pr}[A_i \text{ produces } c \text{ on } f] \cdot \varphi(c) \\ &= \sum_{c \in Y^*} \varphi(c) \cdot \sum_f \mathbf{Pr}[A_i \text{ produces } c \text{ on } f].\end{aligned}$$

By Theorem 3.1, the inner sum is algorithm-independent, so both sides are equal. $\square \quad \square$

Remark. The power $|Y|^{|X|-m}$ is the number of cost functions consistent with any fixed length- m nodup trace: the values on the m queried points are fixed, and the values on the remaining $|X| - m$ points are free. The proof exploits the fact that in a non-revisiting run, every trace is automatically nodup.

4 Theorem 3: Batch Stochastic NFL

4.1 Setting

A *deterministic (batch) learner* A is a function

$$A : \text{List}(X \times \{0, 1\}) \rightarrow (X \rightarrow \{0, 1\})$$

that maps a labelled training sample to a hypothesis. The 0 - 1 loss of hypothesis h on example (x, y) is $\mathbf{1}[h(x) \neq y]$. Given a measure D on $X \times \{0, 1\}$, the *risk* of h is $\text{risk}(D, h) = \int \mathbf{1}[h(x) \neq y] dD(x, y)$.

The *failure probability* of A under D with m i.i.d. samples and risk threshold ε is:

$$\text{failureProb}(D, A, m, \varepsilon) = \mathbf{Pr}_{S \sim D^m}[\text{risk}(D, A(S)) \geq \varepsilon].$$

4.2 Statement

Theorem 4.1 (Batch stochastic NFL [2]). *For any deterministic learner A on a finite domain X (with X nonempty) and any m with $2m \leq |X|$, there exists a distribution D on $X \times \{0, 1\}$ such that:*

1. (Realizability) *Some target $f^* : X \rightarrow \{0, 1\}$ achieves zero risk under D .*
2. (Hardness) $\text{failureProb}(D, A, m, 1/8) \geq 1/7$.

4.3 Proof

Special case $m = 0$. With no training data, the learner A outputs a fixed hypothesis $h_0 = A(\square)$. Pick any $x_0 \in X$ and set $b = \neg h_0(x_0)$. Let $D = \delta_{(x_0, b)}$ (Dirac mass).

1. The constant- b target has zero risk under D .
2. The only sample S of size 0 is the empty tuple; $A(\square) = h_0$; $\text{risk}(D, h_0) = \mathbf{1}[h_0(x_0) \neq b] = 1 \geq 1/8$. Hence $\text{failureProb}(D, A, 0, 1/8) = 1 \geq 1/7$.

Case $m \geq 1$. We work in three steps.

Step 1: Fixed-sample lower bound on average risk.

Choose $C \subseteq X$ with $|C| = 2m$ (possible since $2m \leq |X|$). Let $Z = \{\text{elements of } C\}$.

For a labelling $f : Z \rightarrow \{0, 1\}$ and a sample $S = (z_1, \dots, z_m)$ of m points from Z , define the *abstract risk*:

$$R(f, S) = \frac{1}{|Z|} \sum_{z \in Z} \mathbf{1}[A(\text{sample})(z) \neq f(z)],$$

where $\text{sample} = [(z_1, f(z_1)), \dots, (z_m, f(z_m))]$.

Let $\text{seen}(S) = \{z_1, \dots, z_m\}$ and $\text{unseen}(S) = Z \setminus \text{seen}(S)$. Since $|\text{seen}(S)| \leq m$ and $|Z| = 2m$, we have $|\text{unseen}(S)| \geq m$.

For any *unseen* point $z \notin \text{seen}(S)$, the label $f(z)$ does not appear in the training sample. Therefore the hypothesis $A(\text{sample})$ is the same regardless of the value of $f(z)$. Summing the loss at z over both values of $f(z)$ (while holding the rest of f fixed) gives exactly 1. Summing over all $2^{|Z|-1}$ choices of the other labels:

$$\sum_{f: Z \rightarrow \{0,1\}} \mathbf{1}[A(\text{sample})(z) \neq f(z)] = 2^{|Z|-1} = \frac{2^{|Z|}}{2}.$$

Therefore, for any fixed sample S :

$$\begin{aligned} \sum_{f: Z \rightarrow \{0,1\}} R(f, S) &= \frac{1}{|Z|} \sum_{z \in Z} \sum_{f: Z \rightarrow \{0,1\}} \mathbf{1}[A(\text{sample})(z) \neq f(z)] \\ &\geq \frac{1}{|Z|} \sum_{z \in \text{unseen}(S)} \frac{2^{|Z|}}{2} \\ &\geq \frac{|\text{unseen}(S)|}{|Z|} \cdot \frac{2^{|Z|}}{2} \\ &\geq \frac{m}{2m} \cdot \frac{2^{|Z|}}{2} = \frac{2^{|Z|}}{4}. \end{aligned}$$

Step 2: Contradiction argument.

Suppose for contradiction that every labelling $f : Z \rightarrow \{0, 1\}$ has abstract failure rate $< 1/7$, i.e., the fraction of samples S on which $R(f, S) \geq 1/8$ is less than $1/7$.

For a fixed f , let $B_f = \{S : R(f, S) \geq 1/8\}$ be the “bad” samples. Using $R(f, S) \leq 1$ always and $R(f, S) < 1/8$ for good samples:

$$\begin{aligned} \sum_S R(f, S) &\leq \sum_{S \in B_f} 1 + \sum_{S \notin B_f} \frac{1}{8} \\ &= |B_f| + \frac{|\Omega| - |B_f|}{8} \\ &= \frac{|\Omega|}{8} + \frac{7}{8} |B_f|, \end{aligned}$$

where Ω is the set of all samples of size m from Z . If $|B_f|/|\Omega| < 1/7$, then $|B_f| < |\Omega|/7$, so:

$$\sum_S R(f, S) < \frac{|\Omega|}{8} + \frac{7}{8} \cdot \frac{|\Omega|}{7} = \frac{|\Omega|}{8} + \frac{|\Omega|}{8} = \frac{|\Omega|}{4}.$$

Summing over all $2^{|Z|}$ labellings f and swapping the order of summation:

$$\sum_f \sum_S R(f, S) = \sum_S \sum_f R(f, S) \geq |\Omega| \cdot \frac{2^{|Z|}}{4}.$$

But if every f satisfies the failure-rate assumption:

$$\sum_f \sum_S R(f, S) < 2^{|Z|} \cdot \frac{|\Omega|}{4} = |\Omega| \cdot \frac{2^{|Z|}}{4}.$$

This is a strict inequality contradicting the lower bound. Therefore, some $f^* : Z \rightarrow \{0, 1\}$ must have abstract failure rate $\geq 1/7$.

Step 3: Constructing the hard distribution.

Extend $f^* : Z \rightarrow \{0, 1\}$ to a full target $f_X : X \rightarrow \{0, 1\}$ by setting $f_X(x) = f^*(x)$ for $x \in C$ and $f_X(x) = 0$ otherwise. Let D be the uniform distribution on the labelled graph:

$$D = \frac{1}{|C|} \sum_{x \in C} \delta_{(x, f_X(x))}.$$

1. **Realizability.** The target f_X itself achieves zero 0–1 loss on every point in the support of D , so $\text{risk}(D, f_X) = 0$.
2. **Failure probability.** The risk of any hypothesis h under D equals the empirical risk on C : $\text{risk}(D, h) = \frac{1}{|C|} \sum_{x \in C} \mathbf{1}[h(x) \neq f_X(x)] = R(f^*, \cdot)$. Samples drawn from D^m supported on the graph of f_X correspond (up to an injective embedding) to support-indexed samples. The failure probability under D^m is therefore at least the abstract failure rate $\geq 1/7$.

This completes the proof. $\square$

Remark (Why the constants $1/7$ and $1/8$?). The argument uses the identity $\frac{7}{8} \cdot \frac{1}{7} = \frac{1}{8}$. This ensures that the upper bound $\frac{|\Omega|}{8} + \frac{7}{8} |B_f|$, evaluated at the threshold $|B_f| = |\Omega|/7$, gives exactly $|\Omega|/4$ — matching the lower bound from the unseen-point counting argument. Any pair (ε, δ) with $\varepsilon < 1/2$ and $(1 - \varepsilon)\delta < \varepsilon$ (i.e. $\delta < \varepsilon/(1 - \varepsilon)$) would yield analogous constants.

References

- [1] D. H. Wolpert and W. G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.
- [2] S. Shalev-Shwartz and S. Ben-David. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014.